FIITJEE

ALL INDIA TEST SERIES

CONCEPT RECAPITULATION TEST – II

JEE (Main)-2021

TEST DATE: 17-02-2021

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 Questions.

  Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

  Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer. There is no negative marking.*

Physics

PART – I

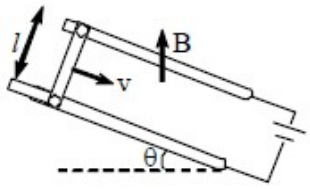
SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. By supplying constant current, the deflection in a moving coil galvanometer falls from 50 to 10 divisions when a shunt of 12 Ω is connected across it. The resistance of the galvanometer coil is
   (A) 24 Ω
   (B) 36 Ω
   (C) 48 Ω
   (D) 60 Ω

2. X-ray from a tube with a target A of atomic number Z shows a K_α lines for target A and two other K_α lines for impurities. The wavelength of K_α lines is λ for target A and λ_1 and λ_2 for two impurities respectively. If $\frac{\lambda}{\lambda_1}=4,\frac{\lambda}{\lambda_2}=\frac{1}{4}$ and the screening constant of Kα lines in unity, select the correct alternative.
   (A) The atomic number of first impurity is 2Z – 1
   (B) The atomic number of first impurity is 2Z + 1
   (C) The atomic number of second impurity is $\frac{Z}{2}+1$
   (D) none of these

3. Two conducting rails are connected to a source of emf and form an incline as shown. A bar of mass m slides without friction down the incline through a vertical magnetic field B. If the length of the bar is l and current i is provided by battery, then the value of B, which will make the bar slide at a constant velocity is



   (A) $\frac{mg\sin\theta}{i\ell}$
   (B) $\frac{mg\cos\theta}{i\ell}$
   (C) $\frac{mg\tan\theta}{i\ell}$
   (D) $\frac{mg}{i\ell\cos\theta}$

4. A p-n-p transistor is used as an amplifier in the common-base configuration. If the ratio of change in current in emitter and corresponding change in current in collector is 1.25, then the value of current gain is :
   (A) 0.8
   (B) 0.9
   (C) 1.0
   (D) 1.25

5. A ball of radius R carries a positive charge whose volume charge density depends only on separation r from the ball's centre as $\rho = \rho_0\left(1-\frac{r}{R}\right)$ where ρ_0 is a constant. Assuming the relative permittivities of the ball and the environment to be equal to unity. The electric field at distance r_1 from centre (For $r_1 < R$) is

(A) $\frac{\rho_0 r_1}{3\varepsilon_0}\left(1-\frac{3r_1}{4R}\right)$

(B) $\frac{\rho_0 r_1}{3\varepsilon_0}\left(1-\frac{r_1}{R}\right)$

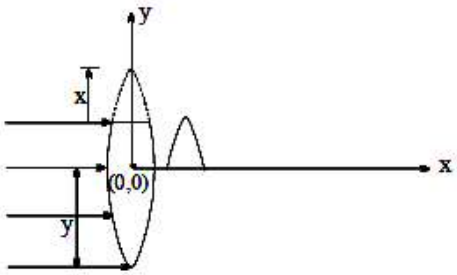
(C) $\frac{\rho_0 r_1}{2\varepsilon_0}\left(1-\frac{r_1}{2R}\right)$

(D) None of these

6. The escape velocity on the surface of the earth is 11.2 km/s. What would be the escape velocity on the surface of another planet of the same mass but 1/4 times the radius of the earth?

(A) 44.8 km/s
(B) 22.4 km/s
(C) 5.6 km/s
(D) 11.2 km/s

7. The dashed part of the lens is cut and kept on the x axis as shown in the diagram. If parallel paraxial rays are falling on this system then the coordinate of image formed after refraction from both the lenses is (30, –1). If x = 2.5 then (all the distance is in cm) (Assume lens have no spherical aberration):



(A) y = 3.5 cm
(B) y = 5.5 cm
(C) y = 3 cm
(D) y = 4.5 cm

8. An infinitely long rod lies along the axis of a concave mirror of focal length f. The near end of the rod is at a distance u > f from the mirror. Its image will have a length

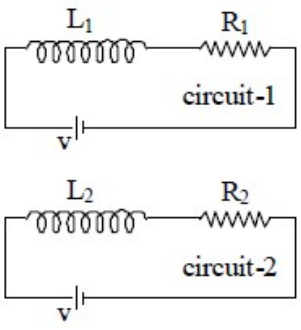
(A) $\frac{f^2}{u-f}$

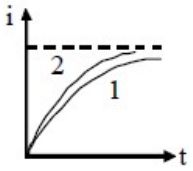
(B) $\frac{uf}{u-f}$

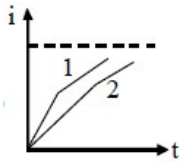
(C) $\frac{f^2}{u+f}$

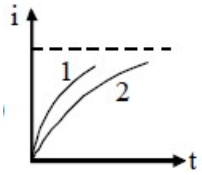
(D) $\frac{uf}{u+f}$

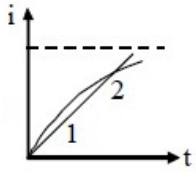
9. For two circuits as shown, find the correct graph between current (i) and time (t), if $L_1 < L_2$: (The steady state current is same for both circuits)



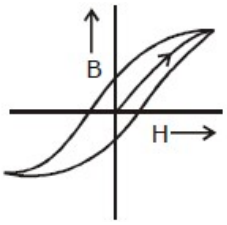
(A) 

(B) 

(C) 

(D) 

10. The B-H curve for a certain specimen is schematically shown by the given diagram. Which one of the following is the correct magnetic nature of the specimen?



(A) Diamagnetic and not ferromagnetic or paramagnetic
(B) Ferromagnetic and not diamagnetic or paramagnetic
(C) Paramagnetic and not diamagnetic or ferromganetic
(D) Applicable to all the three types of magnetism mentioned above

11. In single slit diffraction the linear position of 2nd minima from centre is y, D is screen distance, λ is wavelength, d is slit width then y is

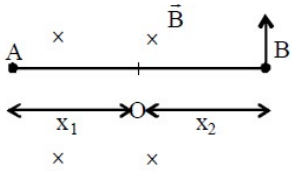
(A) $\frac{d}{\lambda D}$

(B) $\frac{2d}{\lambda D}$

(C) $\frac{2\lambda D}{d}$

(D) $\frac{2Dd}{\lambda}$

12. A metal rod of mass m and length l is rotating with a constant angular velocity ω about an axis passing through O and normal to its length. Potential difference between ends of rod is $\frac{B\omega\ell^2}{6}$, Find x_1 ($x_1 < x_2$) (Magnetic field B is vertically downward) :



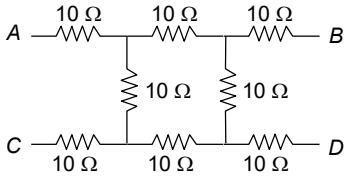
(A) $\frac{\ell}{4}$

(B) $\frac{\ell}{3}$

(C) $\frac{\ell}{6}$

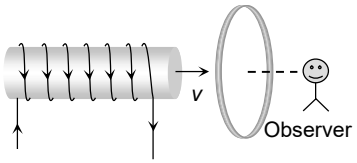
(D) $\frac{\ell}{7}$

13. What will be the equivalent resistance of circuit shown in figure between points *A* and *D*



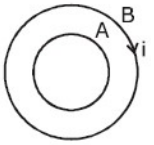
(A) 10 Ω
(B) 20 Ω
(C) 30 Ω
(D) 40 Ω

14. A current carrying solenoid is approaching a conducting loop as shown in the figure. The direction of induced current as observed by an observer on the other side of the loop will be



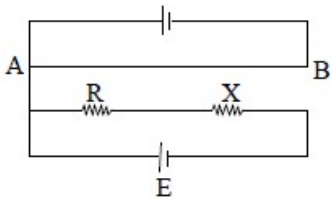
(A) Anticlockwise
(B) Clockwise
(C) East
(D) West

15. If current flowing in outer coil B is given as i = 2t, then induced current in inner coil A will be



(A) Anticlockwise
(B) Clockwise
(C) May be clockwise or Anticlockwise
(D) No induced current

16. In the given figure shows a potentiometer circuit (AB is the potentiometer wire) for comparison of two resistances R and X. A student fails to get the balance point.



Consider the following statements.
(a) Potential difference across the potentiometer wire is greater than that across R and X if one fails to get the balance point
(b) Potential difference across the potentiometer wire is less than that across R and X if one fails to get the balance point
(c) A series resistor is used in the external circuit to get the balance point
(d) A shunt is used in the external circuit to get the balance point
Which of the statements given above are correct
(A) a and c
(B) b and c
(C) a and d
(D) b and d

17. A particle of mass m and carrying charge $-q_1$ is moving around a charge $+q_2$ along a circular path of radius r. Find period of revolution of the charge $-q_1$ about $+q_2$

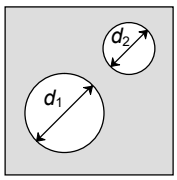
(A) $\sqrt{\frac{16\pi^3\varepsilon_0 mr^3}{q_1q_2}}$

(B) $\sqrt{\frac{8\pi^3\varepsilon_0 mr^3}{q_1q_2}}$

(C) $\sqrt{\frac{q_1q_2}{16\pi^3\varepsilon_0 mr^3}}$

(D) Zero

18. Two holes of unequal diameters d_1 and d_2 $(d_1 > d_2)$ are cut in a metal sheet. If the sheet is heated



(A) Both d_1 and d_2 will decrease
(B) Both d_1 and d_2 will increase
(C) d_1 will increase, d_2 will decrease
(D) d_1 will decrease, d_2 will increase

19. Two liquids *A* and *B* are at 32°*C* and 24°*C*. When mixed in equal masses the temperature of the mixture is found to be 28°*C*. Their specific heats are in the ratio of
(A) 3 : 2
(B) 2 : 3
(C) 1 : 1
(D) 4 : 3

20. A convex lens *A* of focal length 20 *cm* and a concave lens *B* of focal length 5 *cm* are kept along the same axis with the distance *d* between them. If a parallel beam of light falling on *A* leaves *B* as a parallel beam, then distance *d* in *cm* will be
(A) 25
(B) 15
(C) 30
(D) 50

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

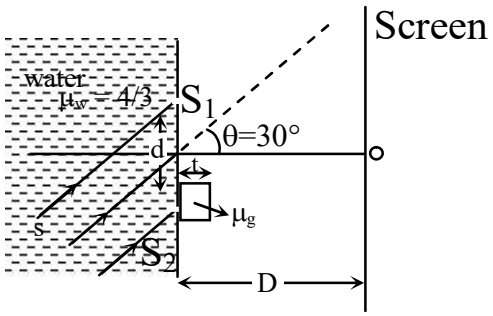
21. A long thin string has a coat of water on it. The radius of the water cylinder is 2 cm. After some time it was found that the string had a series of equally spaced identical water drops on it. Find the minimum distance (in cm) between two successive drops.

22. A railway tank wagon with its 2m diameter and 6m long horizontal cylindrical body, half full of petrol is driven around a curve of radius 100 m, at a speed of 8.33 m/s. The curve runs smoothly into a straight track and the train maintains a constant speed. If value of nf is 2.3. Find the value of 'n'. neglect viscosity and consider petrol as a solid semi cylinder sliding inside the tank. Given: $\tan^{-1}(0.07) \approx 4°$. (f is angular frequency of oscillations)

23. Light of wavelength 1800 Å ejects photoelectrons from a plane of a metal whose work function is 2 eV. If a uniform magnetic field of 5×10^{-5} tesla is applied parallel to plate, what would be the radius of the path (in metre) followed by electrons ejected normally from the plate with maximum energy.

24. Electromagnetic radiation whose electric component varies with time as $E = C_1(C_2 + C_3\cos\omega t)\cos\omega_0 t$, here C_1, C_2 and C_3 are constants, is incident on lithium and librates photoelectrons. If the kinetic energy of most energetic electrons be 0.592×10^{-19} J. Given that $\omega_0 = 3.6 \times 10^{15}$ rad/sec and $\omega = 6 \times 10^{14}$ rad/sec. Find out the work function of lithium in eV (take planks constant $h = 6.6 \times 10^{-34}$ MKS).

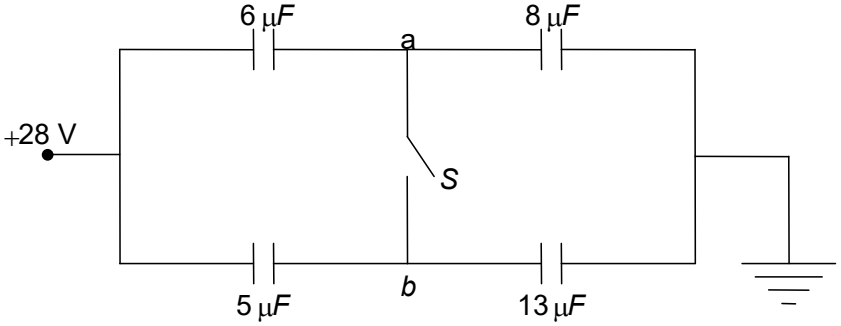
25. In a YDSE experiment, the two slits are covered with a transparent membrane of negligible thickness which allows light to pass through it but does not allow water. A glass slab of thickness t = 0.41 mm and refractive index μ_g = 1.5 is placed in front of one of the slits as shown in the figure. The separation between the slits is d = 0.30 mm. The entire space to the left of the slits is filled with water of refractive index μ_w = 4/3.



A coherent light of intensity I and absolute wavelength λ = 5000A° is being incident on the slits making an angle 30° with horizontal. If screen is placed at a distance D = 1m from the slits, then find the distance of central maxima from O (in cm).

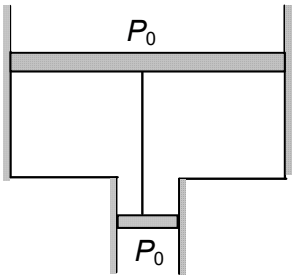
26. A ball of mass 10^{-2} kg and having charge 3×10^{-6} C is tied at one end of a 1 m long thread. The other end of the thread is fixed and a charge -3×10^{-6} C is placed at this end. The ball can move in the circular orbit of radius 1 m in the vertical plane. Initially, the ball is at the bottom. Find the minimum initial horizontal velocity of the ball so that it will be able to complete the full circle. Give answer in SI units.

27. The capacitors in the figure are initially uncharged and are connected as shown.



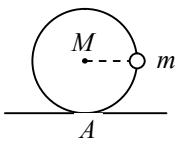
How much charge flowed through the switch when it was closed? (μC)

28. A smooth vertical tube having two different sections is open to the atmosphere at both ends and is equipped with two pistons of corresponding areas; each piston sliding within a respective tube section. One mole of ideal gas is enclosed between the pistons tied with an inextensible thread. The cross-sectional area of the upper piston is 10 cm^2 greater than that of the lower one. The combined mass of two pistons is equal to 5 kg. The outside air pressure p_0 = 1.0 atm. Through what temperature range must the gas between the pistons be heated to shift the pistons through 5.0 cm? Give answer in SI units



29. A U tube contains water and methylated spirit separated by mercury. The mercury columns in the two arms are at the same level with 10.0 cm of water in one arm and 12.5 cm of spirit in the other. What is the relative density of spirit?

30. A uniform body of mass *M* of radius *R* has a small mass *m* attached at edge as shown in the figure. The system is placed on a perfectly rough horizontal surface such that mass *m* is at the same horizontal level as the centre of body. It is assumed that there is no slipping at point *A*. If I_A is the moment of the inertia of combined system about point of contact *A* then the normal reaction at point *A* just after the system is released from rest is 10*n* in SI units find the value of '*n*'.(*M* = 6 kg, *m* = 2 kg, I_A = 4kg m^2, *R* = 1m, g = 10m/s^2)



Chemistry

PART – II

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. Identify the correct order of wavelength of light absorbed by the following complex ions.

$[Co(H_2O)_6]^{3+}$; $Co(CN)_6]^{3-}$;
I II

$[Co(F)_6]^{3-}$; $[Co(en)_3]^{3+}$
III IV

(A) III > I > IV > II
(B) II > IV > I > III
(C) III > I > II > IV
(D) None of these

32. Ratio of $\Delta T_b/K_b$ for 6% (W/V) AB_2 is 1 mol/kg and also for 9% (W/V) A_2B, it is 1 mol/kg (AB_2 and A_2B both are non-electrolytes). Hence, atomic masses of A and B are respectively :
(assume molality = molarity)
(A) 60, 90
(B) 40, 40
(C) 40, 10
(D) 10, 40

33. A hydrogen electrode is immersed in a solution with pH = 0 (HCl). By how much will the potential (reduction) change if an equivalent amount of NaOH is added to the solution. (Take $p_{H_2} = 1$ atm & T = 298K).
(A) increase by 0.41 V
(B) increase by 59 mV
(C) decrease by 0.41 V
(D) decrease by 59 mV

34. Second ionization energy (IE_2/kJ mol^{-1}) order of group 15 elements
(A) N > P > Sb > Bi
(B) P > N > Sb > Bi
(C) N > P > Bi > Sb
(D) N < P < Sb < Bi

35. Which of following is correct set ?

	Species	Hybridisation	Shape
(A)	XeF_2	sp	Linear
(B)	XeO_2F_2	sp^3	Tetrahedral
(C)	$XeOF_4$	sp^3d^2	Square
(D)	XeO_3	sp^3d	Trigonal pyramidal

36. Bromine water reacts with SO_2 to form :
(A) H_2O and HBr
(B) HBr and S
(C) H_2SO_4 and HBr
(D) S and H_2O

37. The number of moles of $KMnO_4$ that will be needed to react completely with one mole of ferrous oxalate in acidic solution is :

(A) $\frac{3}{5}$

(B) $\frac{2}{5}$

(C) $\frac{4}{5}$

(D) 1

38. To neutralize completely 20 mL of 0.1 M aqueous solution of H_3PO_3, the molarity of 40 mL KOH solution required is :

(A) 0.01
(B) 0.05
(C) 0.1
(D) 0.2

39. A radioactive isotope ${}_{Z}X^{A}$ is converted into ${}_{Z-8}Y^{A-16}$ by α-decay. If $t_{50\%}$ is 34.64 hour then disintegration constant of ${}_{Z}X^{A}$ is (in hr^{-1}) is:

(A) 8×10^{-2}

(B) 4×10^{-2}

(C) 2×10^{-2}

(D) 3.6×10^{-3}

40. For cubic structure of diamond:

(A) Effective number of atoms present in a diamond cubic cell is 8.
(B) Each carbon is surrounded by four nearest carbon atoms.
(C) Atomic radius of carbon is $\frac{\sqrt{3}\times a}{8}$. Where a is edge length of cube.
(D) Packing fraction of unit cell is $\frac{\sqrt{3}\times\pi}{8}$.

Select the correct option :

(A) A, B and D are correct
(B) B, C and D are correct
(C) B and C are correct
(D) A, B, C and D are correct

41. Thermal decomposition of $Ba(N_3)_2$ and NaN_3 gives

(A) N_3^-

(B) NH_3

(C) N_2

(D) $NaNH_2$

42. Find the freezing point of a solution when 5 mol of KI is mixed with 1 mol of $HgCl_2$ in 3500 gram of H_2O K_f of $H_2O = 1.86\frac{K}{m}$. $(KI + HgCl_2 \rightarrow K_2HgI_4 + KCl)$

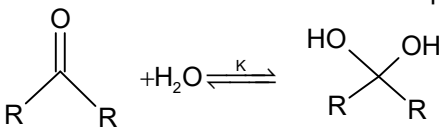
(A) –28.7ºC
(B) –41.45ºC
(C) –35.07ºC
(D) –4.78ºC

43. A quantity of 5 mole of an ideal gas initially at a pressure of 2.0 atm and 400 K is allowed to expand isothermally until its final pressure is 0.5 atm. Calculate the value of the work done if the expansion is carried out against a vacuum?
(A) 1
(B) 2
(C) 3
(D) 0

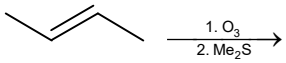
44. Huckel's rule: Planar, fully conjugated monocyclic system with (4n+2n) π electrons have a closed shell of electron, all in bonding orbital's are exceptionally stable such systems are said to be aromatic. Choose incorrect one?
(A) [18] – annulene is aromatic
(B) Benzene is aromatic
(C) All–cis–[10] annule is aromatic
(D) Naphthalene is aromatic

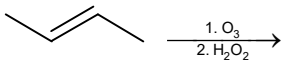
45. 0.2% solution of phenol acts as _____ & 1% solution of phenol acts as _____
(A) antiseptic, disinfectant
(B) disinfectant, antiseptic
(C) analgesic, disinfectant
(D) antipyretic, antiseptic

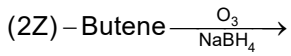
46. Choose the correct about forward equilibrium constant (K) for given set – up?

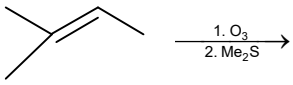


(A) acetaldehyde > acetone
(B) Chloral < acetone
(C) Formaldehyde < acetone
(D) Hexafluoro acetone < acetone

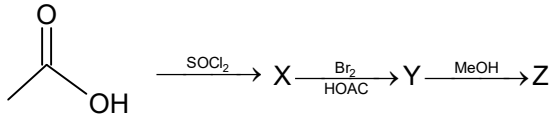
47. Alcohol is obtained from
(A) 

(B) 

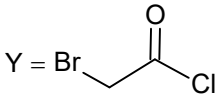
(C) 

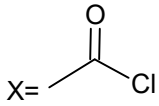
(D) 

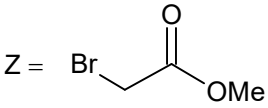
48. Enolizable carbonyl compound is
(A) Benzophenone
(B) Benzaldehyde
(C) Formaldehyde
(D) Cyclopentane carbaldehyde

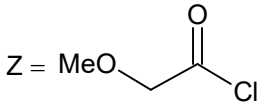
49. 

Choose incorrect X, Y and Z

(A) 

(B) 

(C) 

(D) 

50. Consider the following reactions

(I) D-Glucose $\xrightarrow{NaBH_4}$ X

(II) D-Mannose $\xrightarrow{NaBH_4}$ Y

(III) D-Fructose $\xrightarrow{NaBH_4}$ Products

The product/s in reaction (III) is / are

(A) X only
(B) Y only
(C) Neither X nor Y
(D) Both X and Y

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

51. A buffer solution that is 0.45 M in NH_4Cl and 0.15 M in NH_3. The pH of a given buffer solution is? Given: K_b for ammonia is 3×10^{-5}.

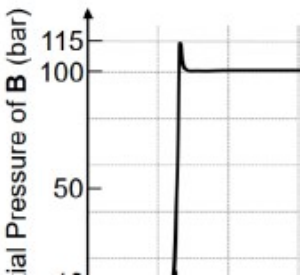
52. How many Natural Occurring Essential α-amino acids has aromatic ring in side chain?

53. In acidic solution, the disaccharide is converted to a mixture of the monosaccharides in a pseudo first order reaction. Under certain conditions of pH, the half-life of sucrose is 28.4 min. To calculate how long (in min.) it takes for the concentration of a sample to fall from 16.0 mmol dm^{-3} to 1.0 mmol dm^{-3}.

54. What is the value of $PK_b(CH_3COO^-)$, if $\lambda_m^\infty = 390$ and $\lambda_m = 7.8$ for 0.4 M of a CH_3COOH solution at 25°C?

55. Reaction of phenol with NaOH followed by heating with CO_2 under high pressure, and subsequent acidification gives compounds X as the major product, which can be purified by steam distillation. When reacted with acetic anhydride in the presence of a trace amount of conc. H_2SO_4 compound X produces Y as the major product. If molecular mass of the final product (Y) is M then report your answer as M/10.

56. Consider the reaction $A \rightleftharpoons B$ at 1000 K. At time t', the temperature of the system was increased to 2000 K and the system was allowed to reach equilibrium. Throughout this experiment the partial pressure of A was maintained at 1 bar. Given below is the plot of the partial pressure of B with time. What is the ratio of the standard Gibbs energy of the reaction at 1000 K to that at 2000 K?



57. Consider a 70% efficient hydrogen-oxygen fuel cell working under standard conditions at 1 bar and 298 K. Its cell reaction is

$$H_2(g)+\frac{1}{2}O_2(g)\rightarrow H_2O(l)$$

The work derived from the cell on the consumption of 1.0×10^{-3}mol of $H_2(g)$ is used to compress 1.00 mol of a monoatomic ideal gas in a thermally insulted container. What is the change in the temperature (in K) of the ideal gas ?
The standard reduction potentials for the two half-cells are given below

$$O_2(g)+4H^+(aq.)+4e^-\rightarrow 2H_2O(l), E^o=1.23V,$$

$$2H^+(aq.)+2e^-\rightarrow H_2(g), E^o=0.00V.$$

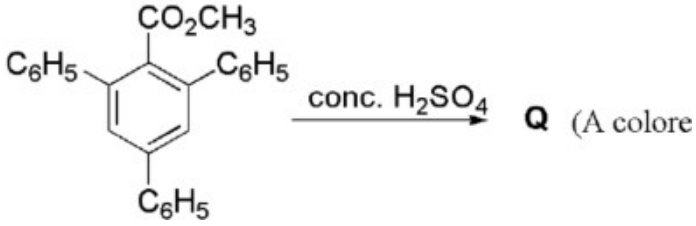
Use F = 96500 C mol^{-1}, $R=8.314\,Jmol^{-1}K^{-1}$

58. Aluminium reacts with sulfuric acid to form aluminium sulfate and hydrogen. What is the volume of hydrogen gas in liters (L) produced at 300 K and 1.0 atm pressure, when 5.4 g of aluminium and 50.0 mL of 5.0 M sulfuric acid are combined for the reaction ?
(Use molar mass of aluminium as 27.0 g mol^{-1}, R = 0.082 atm L mol^{-1} K^{-1})

59. $^{238}_{92}U$ is known to undergo radioactive decay to form $^{206}_{82}Pb$ by emitting alpha and beta particles. A rock initially contained 68×10^{-6}g of $^{238}_{92}U$. If the number of alpha particles that it would emit during its radioactive decay of $^{238}_{92}U$ to in three half-lives is $Z\times10^{18}$, then what is the value of Z?

60. In the following reaction, compound Q is obtained from compound P via an ionic intermediate

What is the degree of unsaturation of Q?



Mathematics

PART – III

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. The angle between the asymptotes of a hyperbola is 30°. The eccentricity of the hyperbola may be
(A) $\sqrt{3} \pm 1$
(B) $\sqrt{3} + 1$
(C) $\sqrt{6} \pm \sqrt{2}$
(D) None of these

62. The chords $lx + my - 1 = 0$ (l, m being parameters) of the curve $x^2 - 3y^2 + 3xy + 3x = 0$, subtending a right angle at the origin, are concurrent at the point
(A) $\left(-\frac{3}{2}, 0\right)$
(B) $\left(\frac{3}{2}, 0\right)$
(C) $\left(\frac{1}{2}, 0\right)$
(D) None of these

63. If foci of hyperbola lie on $y = x$ and one of the asymptote is $y = 2x$, then equation of the hyperbola, given that it passes through (3, 4), is
(A) $x^2 - y^2 - \frac{5}{2}xy + 5 = 0$
(B) $2x^2 - 2y^2 + 5xy + 5 = 0$
(C) $2x^2 + 2y^2 - 5xy + 10 = 0$
(D) none of these

64. If $2x^2 + 2y^2 + 5x - 6y + 1 = 0$ and $xy = 3$ are intersecting at the points (x_i, y_i), i = 1, 2, 3, 4, then
(A) $\sum y_i = +3$, $\prod y_i = 9$
(B) $\sum y_i = 3$, $\prod y_i = 9$
(C) $\sum y_i = -3$, $\prod y_i = -9$
(D) $\sum y_i = 9$, $\prod y_i = 3$

65. Let m_1 and m_2 be the slopes of tangents from the point (–2, 5) on the hyperbola $\frac{x^2}{25} - \frac{y^2}{16} = 1$,
Then the point from which the tangents drawn on the hyperbola have slopes $[m_1]$ and $[m_2]$ and positive intercepts on y-axis is (where [.] denotes the greatest integer function).
(A) (0, 5)
(B) (0, 3)
(C) (3, 0)
(D) (5, 0)

66. Let y be an element of the set A = {1, 2, 3, 5, 6, 10, 15, 30} and x_1, x_2, x_3 be integers such that $x_1 x_2 x_3 = y$, then the number of positive integral solution of $x_1x_2x_3 = y$
(A) 64
(B) 27
(C) 81
(D) none of these

67. If f(X) is a differentiable real valued function satisfying $f''(x) - 3f'(x) > 3, \forall x \geq 0$ and $f'(0) = -1$, then $f(x) + \times, \forall x > 0$ is
(A) Decreasing function of x
(B) Increasing function of x
(C) Constant function
(D) None of these

68. Let $f(x) = \int_0^x \cos\left(\frac{t^2 + 2t + 1}{5}\right) dt, 0 < x < 2, \text{then } f(x)$
(A) Increases monotonically
(B) Decreasing Monotonically
(C) Has one point of local maximum
(D) Has one point of local minimum

69. If $f(x) = \int_0^1 \frac{x^2 + t^2}{2 - t}$ dt, then the curve $y = f(x)$ represents a
(A) Straight line
(B) Parabola
(C) Hyperbola
(D) None of these

70. Area enclosed by curves $x^2 = 4y$ and $y = \frac{8}{x^2 + 4}$ is
(A) $\pi - \frac{2}{3}$
(B) $\pi - \frac{1}{3}$
(C) $2\pi - \frac{4}{3}$
(D) None of these

71. The equation of curve which passes through the point $(0, 0)$ and satisfy the differential equation $\frac{dy}{dx} = \frac{2x + 3y + 1}{y - 3x + 2}$ is
(A) $y^2 + 2x^2 - 6xy + 4y - 2x = 0$
(B) $y^2 - 2x^2 + 6xy + 4y + 2x = 0$
(C) $y^2 - 2x^2 - 6xy + 4y - 2x = 0$
(D) None of these

72. The mean and standard deviation of marks of 200 students were 40 and 15 respectively. Later it was discovered that a score of 40 was wrongly read as 50. The correct standard deviation is
(A) 39.95
(B) 14.98
(C) 224.5
(D) None of these

73. Let $f:R\to R\,f(x)=\dfrac{(x+1)(x-3)}{(x-2)(x-5)}$ then which of following is correct
(A) $f(x)=k$ has two distinct positive if $k>0$
(B) $y=f(x)$ changes into concavity in $\in(2,5)$
(C) $y=f(x)$ increased in some interval
(D) $f(x)$ is into function

74. let $\vec{a}=\hat{i}+\hat{j}+\hat{k},\vec{b}=\hat{i}-\hat{j}+\hat{k},\vec{a}\times\vec{c}=\vec{b}+\lambda\vec{a}$ and $\vec{a}.\vec{c}=1$ then which following is true
(A) $\left[\vec{a}\vec{b}\vec{c}\right]=-\dfrac{8}{3}$ and $\lambda=-\dfrac{1}{3}$
(B) $\left[\vec{a}\vec{b}\vec{c}\right]=\dfrac{8}{3}$ and $\lambda=-\dfrac{2}{3}$
(C) $\left[\vec{a}\vec{b}\vec{c}\right]=-\dfrac{8}{3}$ and $\lambda=-\dfrac{2}{3}$
(D) $\left[\vec{a}\vec{b}\vec{c}\right]=-\dfrac{8}{3}$ and $\lambda=\dfrac{2}{3}$

75. Let O is the centre of the hyperbola $9x^2 - 16y^2 - 144 = 0$. The tangent at any point P on the hyperbola meets the straight lines $3x - 4y = 0$ and $3x + 4y = 0$ at Q and R respectively, then $\sqrt{OQ.OR}$ is
(A) 5
(B) 10
(C) 15
(D) 20

76. The focus of $\dfrac{(y+x)^2}{16}-\dfrac{(y-x)^2}{8}=1$ is at
(A) $\left(\pm\sqrt{6},\pm\sqrt{6}\right)$
(B) $\left(\sqrt{\dfrac{5}{2}},\sqrt{\dfrac{5}{2}}\right)$
(C) $\left(\sqrt{\dfrac{7}{2}},\sqrt{\dfrac{7}{2}}\right)$
(D) None of these

77. A seven digit number of form $a_1a_2a_3a_4a_5a_6a_7$ (all digits distinct) is randomly formed. The probability that number formed satisfy $a_1 > a_2 > a_3 < a_4 < a_5 < a_6 < a_7$, is

(A) $\frac{7}{1512}$

(B) $\frac{11}{1572}$

(C) $\frac{13}{1572}$

(D) $\frac{5}{1512}$

78. If x, y, z are the distance of vertices A, B, C respectively from the orthocentre of ΔABC, then $\frac{xy}{ab}+\frac{yz}{bc}+\frac{zx}{ca}$ is /are

(A) 8
(B) 4
(C) 2
(D) 1

79. If a curve is such that line joining $P(x,y)$ (on curve) to origin with x-axis makes half angle than the angle which tangent at P makes with line drawn from P parallel to y-axis. Then equation of curve may be

(A) $\left(x^2-3y^2\right)x = A$

(B) $\left(x^2+3y^2\right)x = A$

(C) $\left(x^2-y^2\right)x = A$

(D) None of these

80. $(p \wedge q) \wedge (\sim p \wedge q)$ is

(A) A tautology
(B) A contradiction
(C) Both of tautology and a contradiction
(D) Neither a tautology nor a contradiction

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. In a acute angle triangle ABC, the foot of the perpendicular from A divide the opposite side into two parts of length 3 and 17 units and $\tan A = \frac{22}{7}$, the $\left(\frac{\text{Area of } \Delta ABC}{22}\right)$ is

82. If area of square circumscribing the ellipse E is 10 square units and maximum distance of a normal from the centre of ellipse is 1 unit, then $(2e)^2$ is (where e is eccentricity of ellipse)

83. There are n different books each having m copies. If the total number of ways of making a selection of at least one from them is 255 and m – n + 1 = 0, then distance of point (m, n) from the origin is

84. If $\alpha, \beta, \gamma \in R^+$ such that $\alpha + \beta + \gamma = \sqrt{3}$, then minimum value of $\frac{\pi}{\tan^{-1}\alpha + \tan^{-1}\beta + \tan^{-1}\gamma}$ is/are

85. Let $\cot^2\frac{\theta}{2} = \sec\theta - \cos\theta$, then $5\tan^4\frac{\theta}{2}$ is

86. The product of perpendiculars drawn from any point on $\frac{x^2}{9} - \frac{y^2}{16} = 1$ upon its asymptote is

87. If in a triangle ABC, b + c = 4a. Then $\cot\frac{B}{2}\cot\frac{C}{2}$ is equal to

88. If the parabola $y = ax^2 + bx + c$ has vertex at (4, 2) and $a \in [1, 3]$, then difference between the extreme values of abc is equal to,

89. If in a triangle ABC, $a^4 + b^4 + c^4 = 2a^2b^2 + b^2c^2 + 2a^2c^2$, then sin A is equal to

90. The area bounded by $y = \max\{|x-2|+2,\ 3-|x-2|\}$ and $y = \min\{|x-2|+2,\ 3-|x-2|\}$ is